claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde
xer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-
a833930266926a5e7.jsonl`
- SHA-256: `e3969396ff6dbaf84d1c30d4e19025d2b14a4c3ba26fe0187320904767ec6d54`
- Source modified: `2026-06-06T17:34:27+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:33:52.981Z)

Adversarial Claim Verifier (voter 3/3)

```
Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.
```

Research question

```
Research alternative methods to automatically detect rally start/end segments (temporal
segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST
pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach
or whether other methods are competitive or complementary. Keep it focused ("mini" research):
prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA
needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
```

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review
"ShuttleSet is a human-annotated stroke-level badminton singles dataset containing 104 sets, 3,685 rallies, and 36,492 strokes across 44 matches (2018-2021), with stroke-level annotations including 18 shot-type classes, hitting locations, and both players' locations at each stroke."

Source: https://arxiv.org/abs/2306.04948 (primary)
Supporting quote: "104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021"

Checklist
1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:33:55.752Z)

I'll verify this claim about ShuttleSet. Let me start by examining the primary source and searching for corroborating/contradicting evidence.

3. assistant (2026-06-06T17:34:00.221Z)

Let me verify the specific numbers in this claim against the primary source.

4. user (2026-06-06T17:34:05.308Z)

ShuttleSet Dataset Summary

The paper describes **ShuttleSet**, a badminton singles dataset with the following specifications:

Exact Numbers:
- Sets: 104
- Rallies: 3,685
- Strokes: 36,492
- Matches: 44
- Year range: 2018-2021
- Shot-type classes: 18 distinct classes
- Yes, this is singles badminton

Annotations Include:
According to the abstract, annotations encompass "the shot type with a choice of 18 distinct classes, the corresponding hitting locations, and the locations of both players at each stroke." Therefore, the dataset includes both hitting locations and the positions of both players.

Verbatim Quote:
"It contains 104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021

with 27 top-ranking men's singles and women's singles players."

5. user (2026-06-06T17:34:07.054Z)

Web search results for query: "ShuttleSet badminton dataset 104 sets 3685 rallies 36492 strokes 44 matches"

Links: [{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis - ADS","url":"https://ui.adsabs.harvard.edu/abs/2023arXiv230604948W/abstract"},{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | DeepAI","url":"https://deepai.org/publication/shuttleset-a-human-annotated-stroke-level-singles-dataset-for-badminton-tactical-analysis"},{"title":"[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis","url":"https://arxiv.org/abs/2306.04948"},{"title":"(PDF) ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis","url":"https://www.researchgate.net/publication/371413847_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"},{"title":"[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis","url":"https://ar5iv.labs.arxiv.org/html/2306.04948"},{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis - National Yang Ming Chiao Tung University Academic Hub","url":"https://scholar.nycu.edu.tw/en/publications/shuttleset-a-human-annotated-stroke-level-singles-dataset-for-bad/"},{"title":"[PDF] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Semantic Scholar","url":"https://www.semanticscholar.org/paper/ShuttleSet:-A-Human-Annotated-Stroke-Level-Singles-Wang-Huang/405132e19aa07c8a9ae6f8a4e6f440b00a105027"},{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for","url":"https://arxiv.org/pdf/2306.04948"},{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining","url":"https://dl.acm.org/doi/10.1145/3580305.3599906"},{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Request PDF","url":"https://www.researchgate.net/publication/372910678_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"}]

I found extensive information about the ShuttleSet badminton dataset. Here are the search results:

ShuttleSet is the largest publicly-available badminton singles dataset with annotated stroke-level records, containing 104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021 with 27 top-ranking men's singles and women's singles players.

The dataset is manually annotated with a computer-aided labeling tool, with shot type annotations from a choice of 18 distinct classes, along with corresponding hitting locations and the locations of both players at each stroke.

Multiple benchmarks are provided (stroke influence, stroke forecasting, and movement forecasting) with baselines to demonstrate the practicability of using ShuttleSet for turn-based sports analytics.

The dataset was published at the KDD (Knowledge Discovery and Data Mining) conference in 2023 and is designed to support research in sports analytics, tactical analysis, and AI applications for badminton.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:34:18.498Z)

The claim is fully corroborated by the primary source (arXiv abstract) and the KDD 2023 publication. Let me verify the exact wording matches by checking the abstract text directly for any subtle misreads.

The evidence is consistent across the primary arXiv abstract, the ACM SIGKDD 2023 proceedings, and multiple independent indexers (ADS, DeepAI, Semantic Scholar). Every number in the claim matches verbatim. Let me do one more check to confirm there's no qualification I'm missing (e.g., the "human-annotated" and "stroke-level" descriptors, and the singles vs doubles point).

The abstract verbatim states: "It contains 104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021" and "the shot type with a choice of 18 distinct classes, the corresponding hitting locations, and the locations of both players at each stroke." The title itself is "A Human-Annotated Stroke-Level Singles Dataset for Badminton."

Adversarial check complete:
1. **Supported by quote?** Yes ? every figure (104/3,685/36,492/44, 2018-2021) is verbatim in the abstract. The "18 shot-type classes, hitting locations, both players' locations" detail is also verbatim from the abstract's annotation description, not an overreach.
2. **Contradicting evidence?** None found. Multiple independent sources (ACM DL, ADS, Semantic Scholar, DeepAI) report identical numbers.
3. **Source quality?** Strong ? peer-reviewed KDD 2023 paper + primary arXiv abstract. Matches claim strength.
4. **Outdated?** Dataset specs are static historical facts; not subject to drift.
5. **Marketing/cherry-picked?** No ? these are dataset composition facts, independently verifiable.

The claim survives all refutation attempts.

7. user (2026-06-06T17:34:23.497Z)

Structured output provided successfully

8. assistant (2026-06-06T17:34:27.196Z)

The claim is verified and survives all refutation attempts. Every number (104 sets, 3,685 rallies, 36,492 strokes, 44 matches, 2018-2021) and every annotation detail (18 shot-type classes, hitting locations, both players' locations, human-annotated, stroke-level, singles) matches the primary arXiv abstract verbatim and is corroborated by the peer-reviewed KDD 2023 publication plus multiple independent indexers. refuted=false, high confidence.